

The empirical recurrence for OEIS A235445: recovering a conjecture that is not written down

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Abstract

OEIS A235445 records “Empirical recurrence of order 53”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 121$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 53, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 53 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235445 is “Number of $(n+1) \times (4+1)$ 0..1 arrays with the difference between each 2×2 subblock maximum and minimum lexicographically nondecreasing rowwise and columnwise.”. It has offset 1 and begins

742, 15936, 335898, 7017768, 146213244, 3043145826, 63315473350, 1317184566572, ...

The entry states:

Empirical recurrence of order 53 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:34:24, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 583$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 121$ of the 583, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 121$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1166$ exact terms of the model returns order 53 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{53} c_i a(n-i),$$

c_1	33	c_{15}	−111194058	c_{29}	6600765238717	c_{43}	−8928387988
c_2	−222	c_{16}	64053137702	c_{30}	−3588543670672	c_{44}	−1819408044
c_3	−1489	c_{17}	14129649806	c_{31}	−6021549065737	c_{45}	1038277596
c_4	17387	c_{18}	−266367864284	c_{32}	1827600921818	c_{46}	213448380
c_5	23473	c_{19}	−112292275270	c_{33}	4160724563815	c_{47}	−85208568
c_6	−575160	c_{20}	837350478536	c_{34}	−555448547617	c_{48}	−15822784
c_7	−48189	c_{21}	507449727674	c_{35}	−2175937339153	c_{49}	4721472
c_8	11547914	c_{22}	−1991884024230	c_{36}	23599410471	c_{50}	673408
c_9	−3405981	c_{23}	−1548499763906	c_{37}	857849143502	c_{51}	−160512
c_{10}	−158790773	c_{24}	3581611481497	c_{38}	66819455156	c_{52}	−12544
c_{11}	51060507	c_{25}	3386904502729	c_{39}	−253267915896	c_{53}	2560
c_{12}	1577539441	c_{26}	−4846541905470	c_{40}	−35636001040		
c_{13}	−296688600	c_{27}	−5463448584269	c_{41}	55503127732		
c_{14}	−11610564120	c_{28}	4885084679551	c_{42}	10104395700		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 53, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $53 + 4$ terms removed returns order 53 as well, so $\deg q_{\min} = 53 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $53 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 53 that this sequence satisfies — and that family turns out to have exactly one member.

References

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